EDITS:

Enhancing DIgital TwinS with scalable and interpretable AI:

from physics-based modelling and simulation to adaptive learning, big-data analytics and counterfactual reasoning

MLG
MACHINE LEARNING GROUP

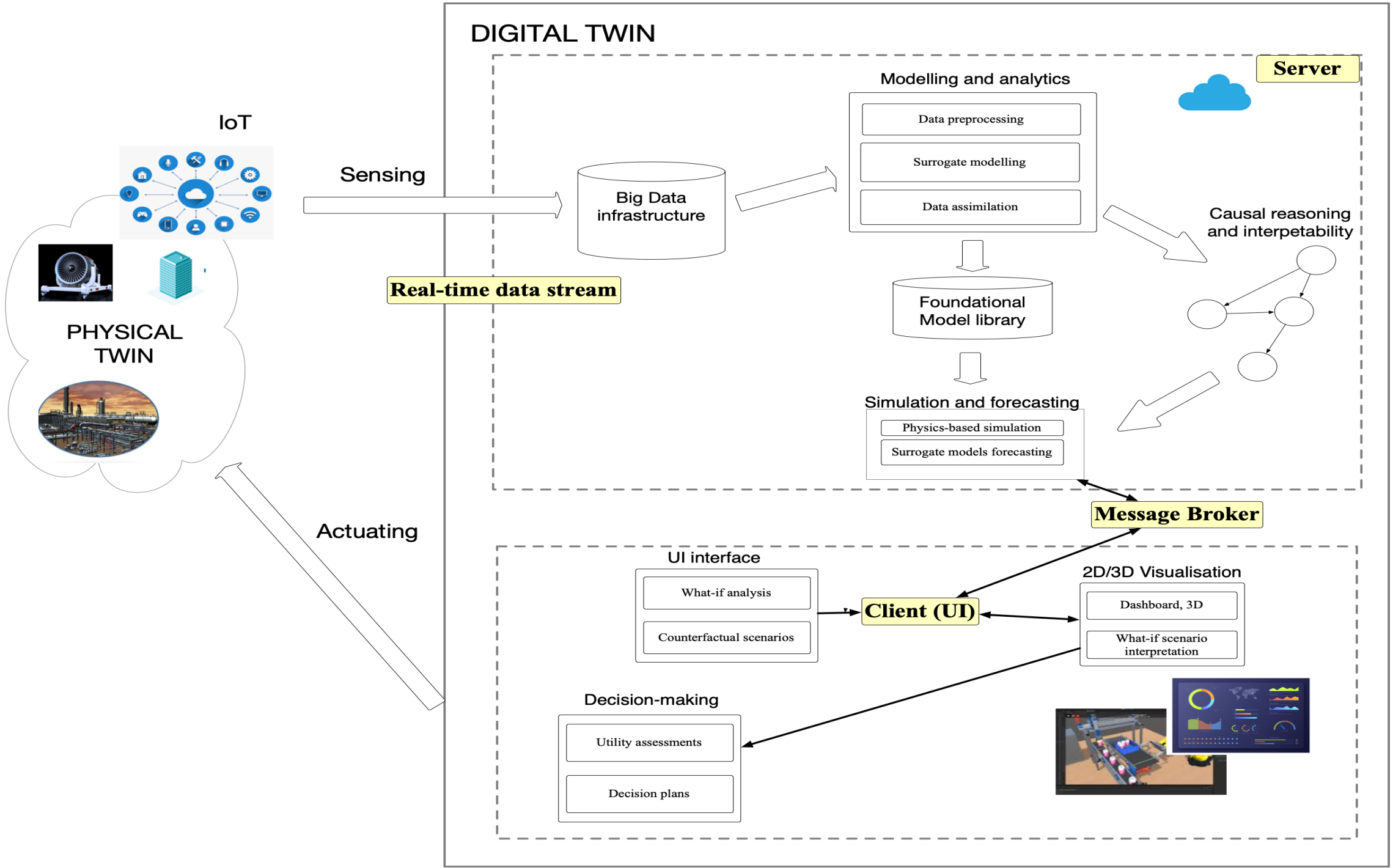
Digital twin (DT)

- Digital avatar of an object/system/process called **physical twin** (PT),
- A DT aims to
  - emulate,
  - explain,
  - anticipate,
  - control

the behaviour of the PT by relying on

  - multiscale physics-based models
  - data-driven ML models

whose accuracy, up-to-dateness and validity are guaranteed by a **continuous two-way flow of data** (measures and actions) between the DT and the PT.


DIGITAL TWIN
Server
IoT
Sensing
Real-time data stream
PHYSICAL TWIN
Actuating
Big Data infrastructure
Modelling and analytics
Data preprocessing
Surrogate modelling
Data assimilation
Foundational Model library
Simulation and forecasting
Physics-based simulation
Surrogate models forecasting
Causal reasoning and interpetability
Message Broker
UI interface
What-if analysis
Counterfactual scenarios
Client (UI)
2D/3D Visualisation
Dashboard, 3D
What-if scenario interpretation
Decision-making
Utility assessments
Decision plans

Beyond the buzzword… hot scientific questions

- Integration of physical and data-driven models
  - Data assimilation
  - Physically based machine learning
  - Surrogate modelling
  - Causal inference and discovery
- Optimisation
  - Multi-criteria
- ML
  - Transfer learning, active learning
  - Online learning
    - Non-stationarity
    - Concept drift
- Data infrastructure
  - big data
  - streaming infrastructure
  - Sequential model updating
  - simulation tools and languages
- User interaction
  - What-if scenario
  - Dashboarding/visualisation

WEL-T case studies

- Toy examples:
  - Plant
  - Robotic arm?
  - ?
- Smart cities
  - Traffic Twin
- Energy
  - Grid
  - Transformers
  - Wind turbines
- Robotic hand

WPs

1. Case studies definition
2. Sensing and acting
   - Sensors
     - Real/virtual
   - Actuators
3. Data Infrastructure
   - Big data streaming:
     - Sampling from data streams (reservoir algorithm)
     - Forgetting
   - Data collection and storage
     - Database
     - Big data architecture
     - Cloud server
4. Orchestration
   - Centralized data
   - Scalability
   - Asynchronous access and training
   - Model storage
   - Kubeflow? MlFlow?
4. Learning
   - Continual/online
   - Batch/offline
5. Data assimilation
   - Integration with physical models
6. Forecasting
7. Simulation
   - What-if scenarios
   - User interaction
   - Visualisation
8. Causality
   - Counterfactuals
   - interpretability
9. Close the loop
   - Decision-making
   - Control
10. Assessment
    - Dashboarding
    - KPIs

Resources

- Gianluca
- Gianmarco: EDITS
- Eladio : EDITS
- Alejandro: MultiRoof
- Cedric: assistant
- Martin: assistant
- Pascal: assistant
- Additional postdocs: NEXUS, EDITS, TORRES2, TreeCity, FARI, POC?
- MA thesis
  - Rayan: MA2
  - Capucine: MA1
  - Centra: MA2 from Italy
  - Interns ?

Next steps

- Who does what?
  - Case study assignment
  - WP assignment
- Federating toy example
- Workgroups
  - Case study (plant, energy, simulation), material
  - Stack

- Presentations/demos:
  - Cloud infrastructure: Gianmarco
  - Orchestration: Kubernetes , Mlflow (Pascal)
  - Isaac Gym library (Cedric)
  - Dashboards

Objectives

- Design a DT methodology
- Beyond fragmented contributions
- Streghten the interaction on common goals
- Convergence of
  - several MLG projects
  - MA thesis
  - Interns topics
- Common infrastructure to be jointly defined/assessed and reused
- Dissemination/ Valorisation/ presentations